

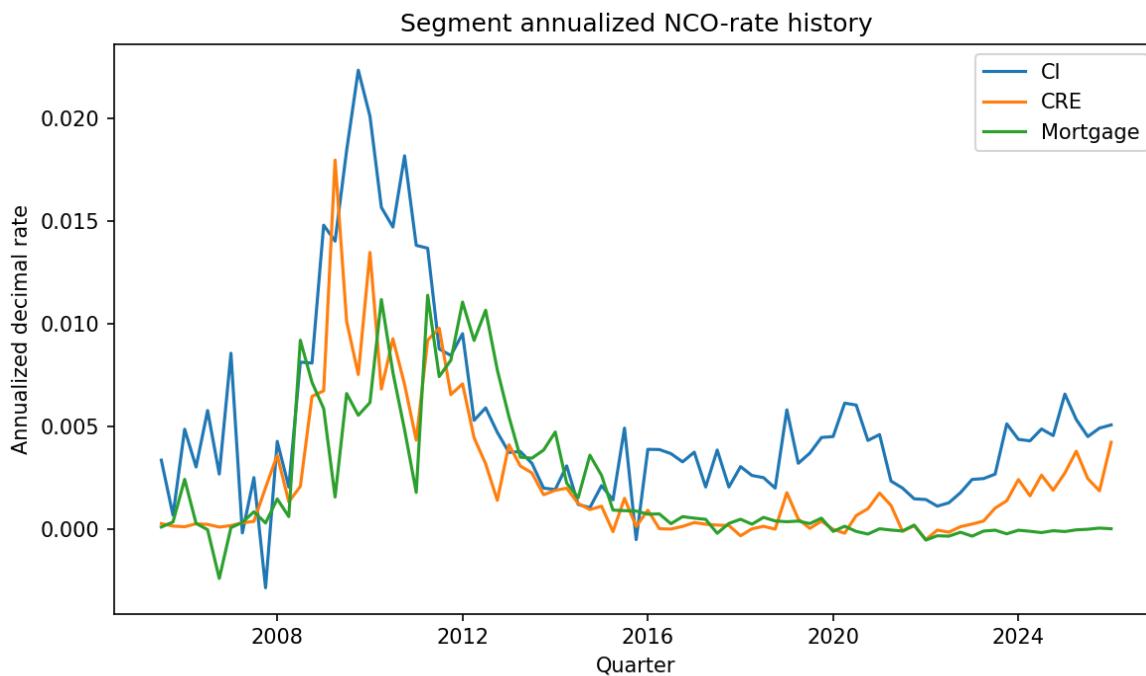
1. Executive summary

Limited R4 closes the repair cycle with 31 common stress banks and three authorized model/segment pairs.

All reported ratios are modeled credit-loss burdens relative to starting Tier1, not capital depletion or CET1 changes.

2. Data and sample

The repaired historical panel contains 33 banks and 8,145 rows. Missing regulatory values remain missing and unsupported reporting intervals are explicit.



3. Data definitions and QA

R1 independently audited effective-dated mappings, component completeness, NCO flow reconstruction, and capital reconciliation. Two bounded flow rows remain review-required.

4. Mean-model evidence

AR is the comparison baseline. Dynamic FE improved equal-observation RMSE in only 1 of 8 R2 segment-window comparisons and is retained in R4 only as a limited C&I; challenger.

5. Tail and Bayesian evidence

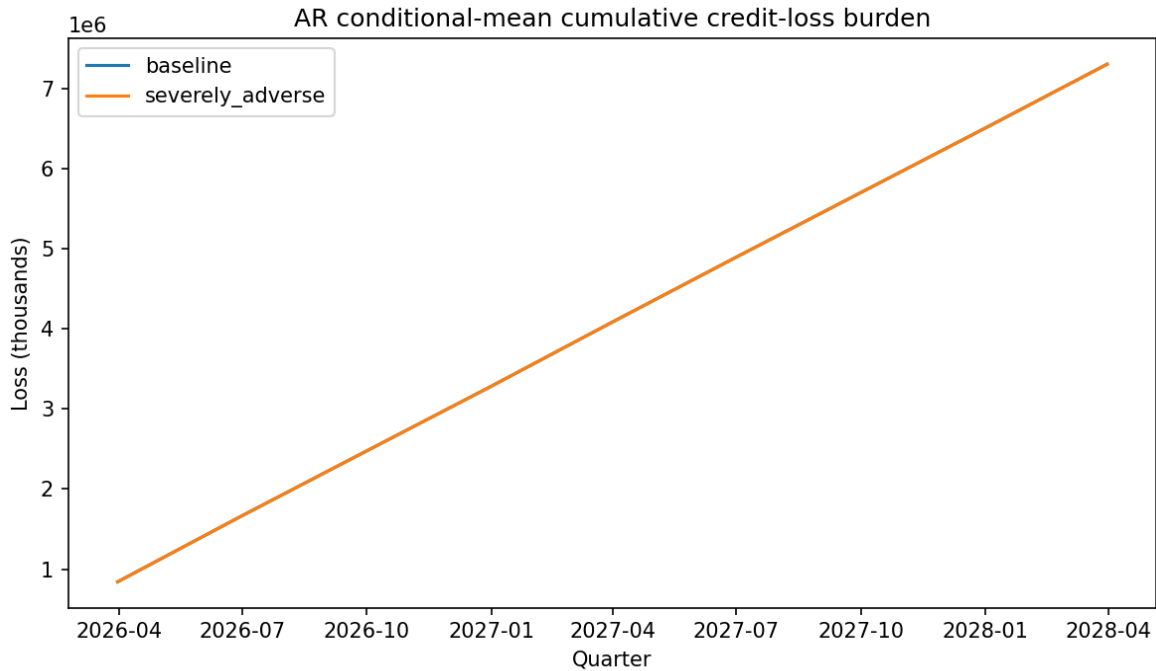
Quantile models have one-step diagnostic use only. No recursive quantile stress paths, cumulative-loss quantiles, or tail rankings are reported. Bayesian and residual-calibration expansion were not evaluated in this bounded stage.

6. Scenario contract

The Federal Reserve 2026 baseline, severely adverse, and historic domestic CSV byte hashes matched the retained manifest. Paths span 2026Q1 through 2028Q1 and preserve macro source quarters, lags, units, and historical boundary lineage.

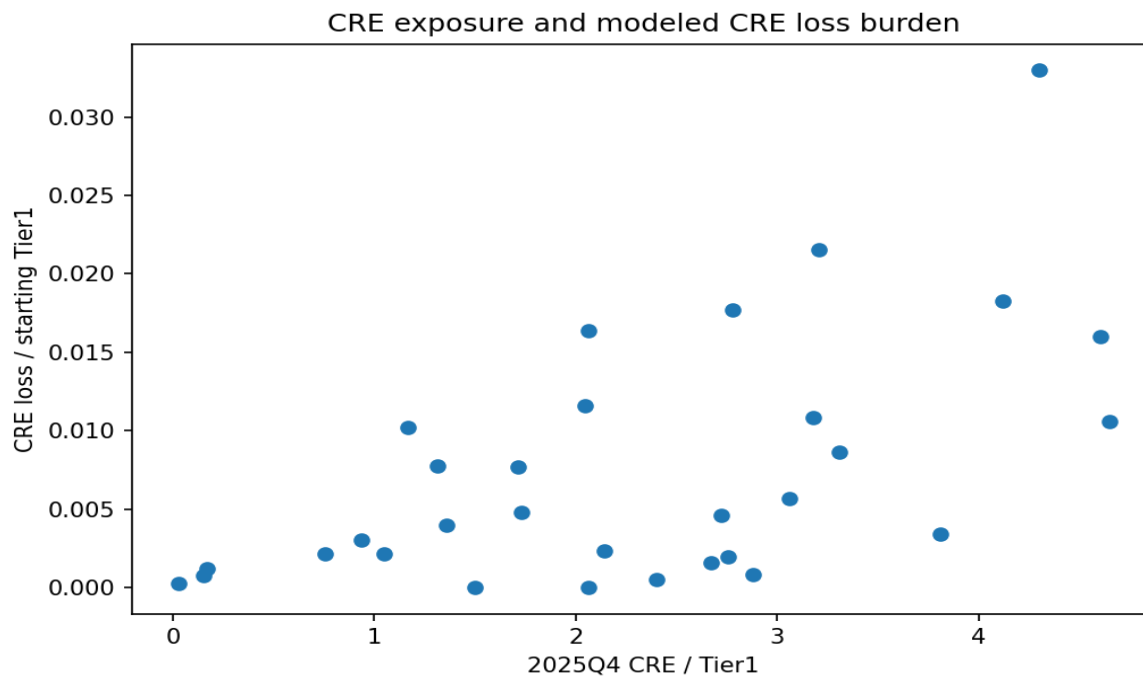
7. Conditional-mean stress results

The AR C&I; plus CRE aggregate is 7,299,383 thousand under baseline and 7,299,383 thousand under severe. AR has no scenario variables, so these totals are identical; this is a baseline-model limitation, not evidence that the scenarios are equally stressful.



8. CRE decomposition

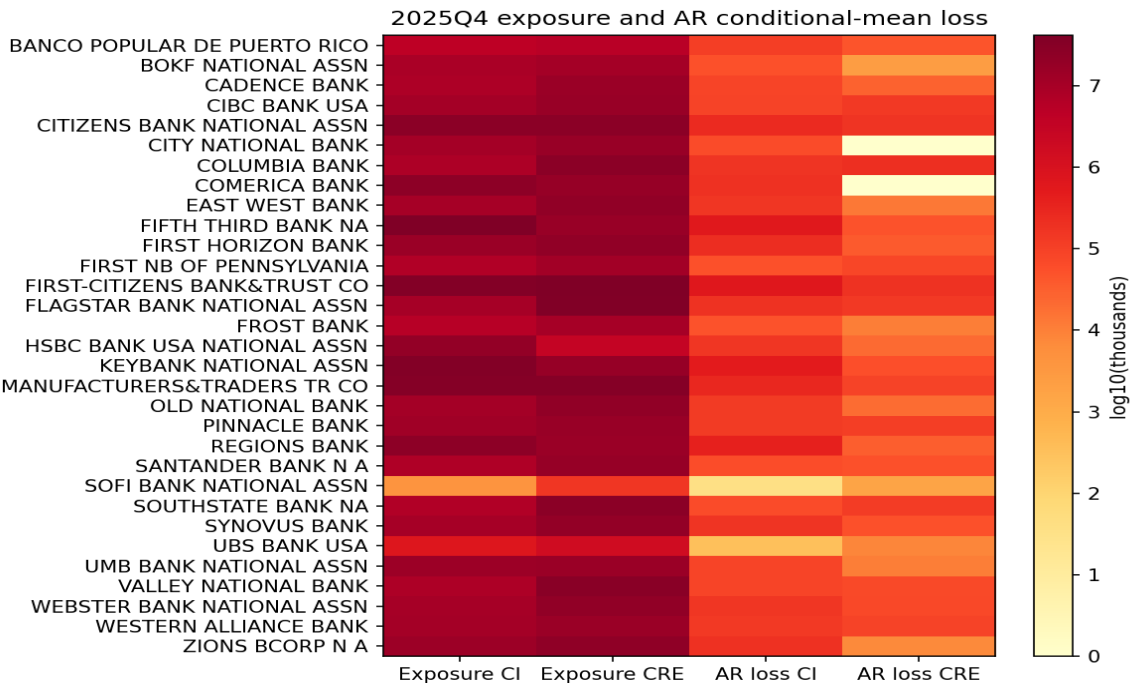
The high-minus-low AR modeled CRE loss-burden difference is 1.02%. The equal-loss-rate comparison is a mechanical exposure/Tier1 decomposition, not a causal concentration estimate. R2 did not support incremental CRE amplification.



9. Sensitivity and reproducibility

The delivered sensitivities cover lambda, partial shocks, exposure changes, the negative-NCO floor, and a common-bank C&I; model comparison.

Unit, integration, artifact-consistency, and full end-to-end evidence are reported separately. The full raw-to-report repair chain was not rerun in this R4 invocation.



10. Research-question disposition

RQ1: INCONCLUSIVE. RQ2: SUPPORTED. RQ3: NOT_SUPPORTED. RQ4: INCONCLUSIVE.

The permitted conclusion is a reproducible, limited conditional-mean stress analysis. A full capital roll-forward, Mortgage stress, multi-period tail distribution, Bayesian model, market validation, and machine-learning expansion remain outside scope.